

EXPERIMENT NO: 12

EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS

AIM:

To write an Embedded C program to multiply two 16-bit numbers using Keil software.

SOFTWARE REQUIRED:

- Keil

PROGRAM:

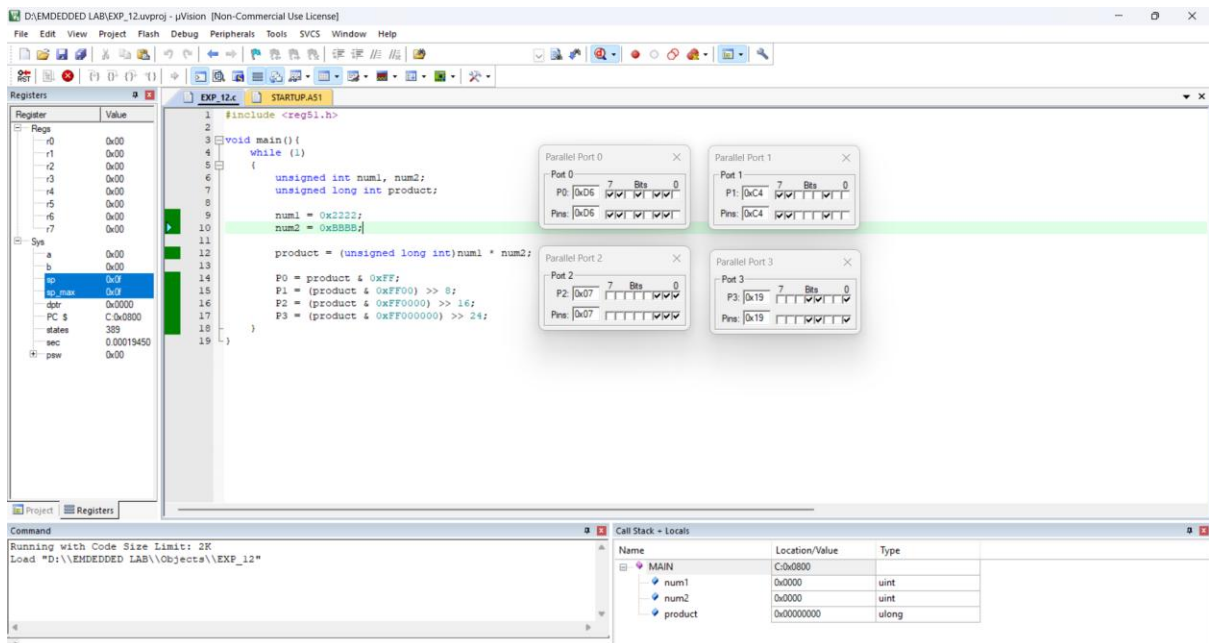
```
#include <reg51.h>

void main(){
    while (1)    {
        unsigned int num1, num2;
        unsigned long int product;
        num1 = 0x2222;
        num2 = 0xBBBB;
        product = (unsigned long int)num1 * num2;
        P0 = product & 0xff;
        P1 = (product & 0xff00) >> 8;
        P2 = (product & 0xff0000) >> 16;
        P3 = (product & 0xff000000) >> 24;    }}

```

The program multiplies the two 16-bit hexadecimal numbers **0x2222** and **0xBBBB** and stores the resulting 32-bit product across **P0, P1, P2, and P3**.

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.